

“Calculus 3”

Multi-Variable Calculus

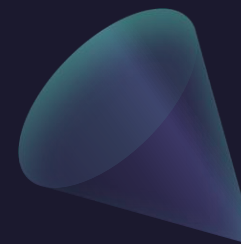
Instructor: Álvaro Lozano-Robledo

Day 24



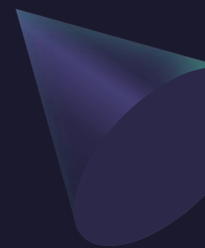
Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.5, 16.6 (curl, divergence, parametric surfaces)
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104





ALVARO: Start the recording!



“Calculus 3”

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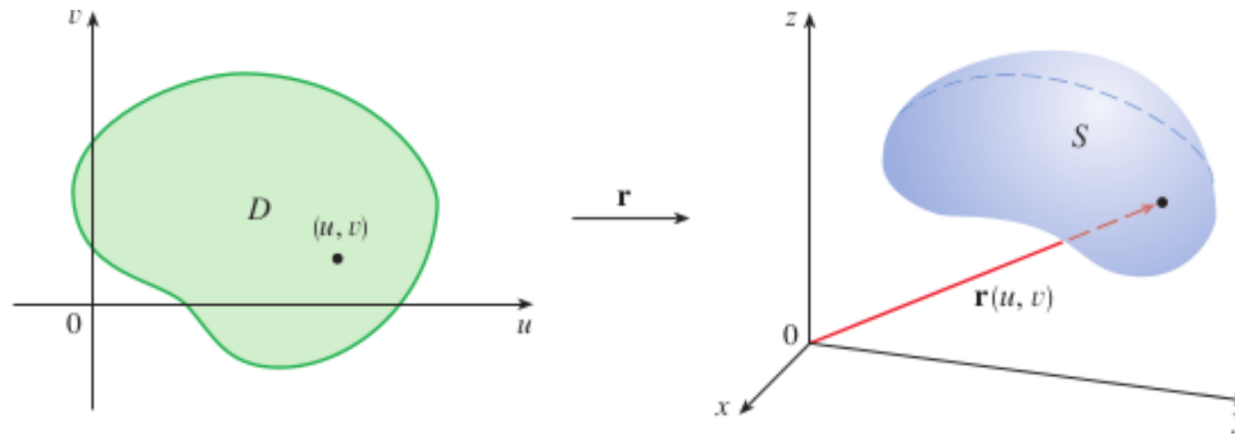
Surface Area



About Parametrized Surfaces and their Area

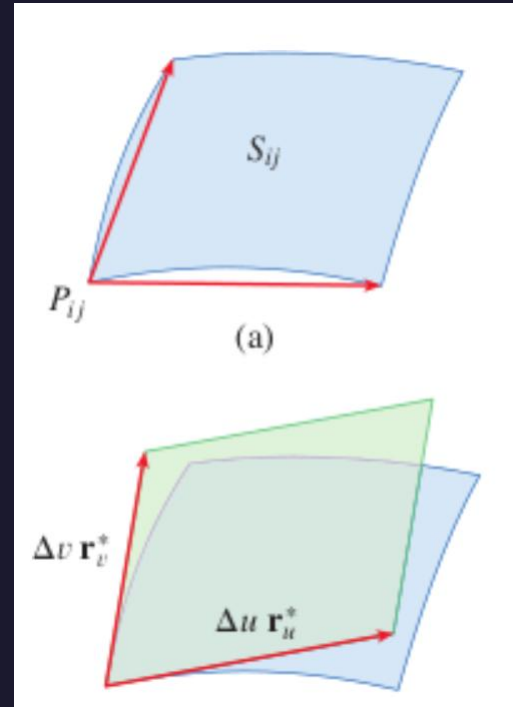
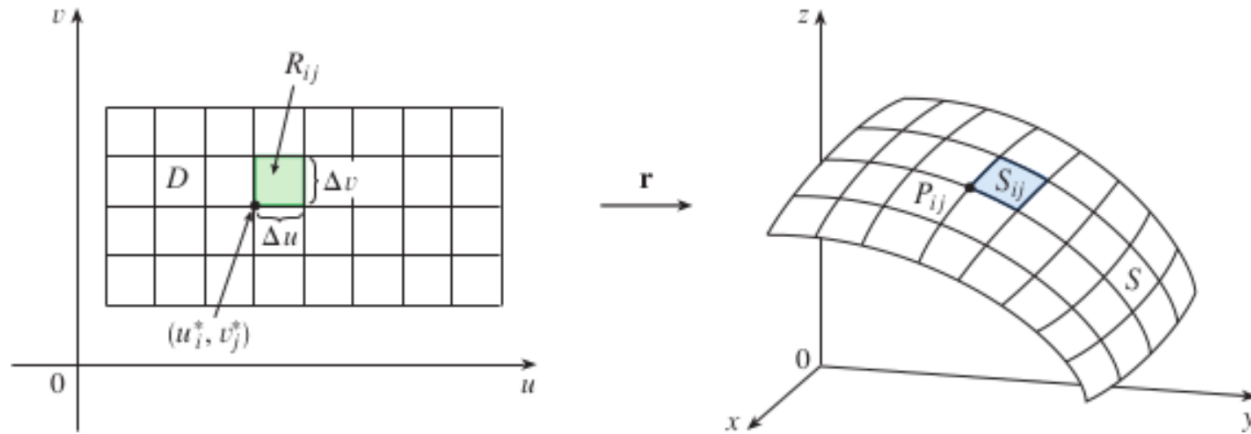
$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

A parametric surface



About Parametrized Surfaces and their Area

The image of the subrectangle R_{ij} is the patch S_{ij} .



$$|(\Delta u \mathbf{r}_u^*) \times (\Delta v \mathbf{r}_v^*)| = |\mathbf{r}_u^* \times \mathbf{r}_v^*| \Delta u \Delta v$$

$$\sum_{i=1}^m \sum_{j=1}^n |\mathbf{r}_u^* \times \mathbf{r}_v^*| \Delta u \Delta v$$

About Parametrized Surfaces and their Area

If a smooth parametric surface S is given by the equation

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k} \quad (u, v) \in D$$

and S is covered just once as (u, v) ranges throughout the parameter domain D , then the **surface area** of S is

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| dA$$

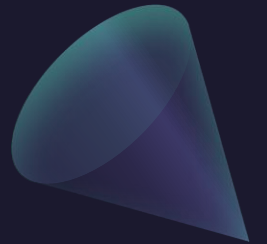
where

$$\mathbf{r}_u = \frac{\partial x}{\partial u} \mathbf{i} + \frac{\partial y}{\partial u} \mathbf{j} + \frac{\partial z}{\partial u} \mathbf{k} \quad \mathbf{r}_v = \frac{\partial x}{\partial v} \mathbf{i} + \frac{\partial y}{\partial v} \mathbf{j} + \frac{\partial z}{\partial v} \mathbf{k}$$

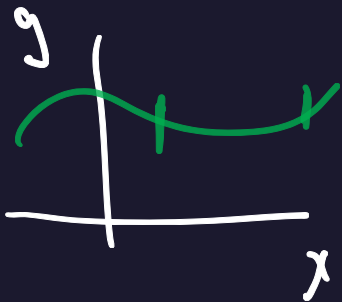
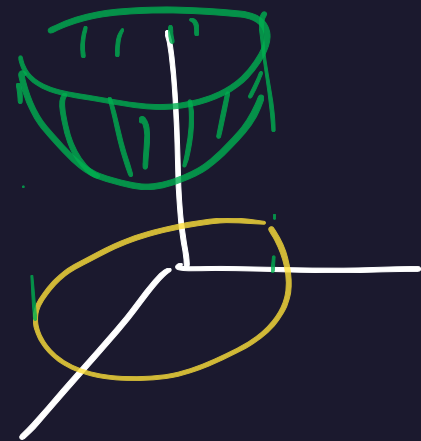
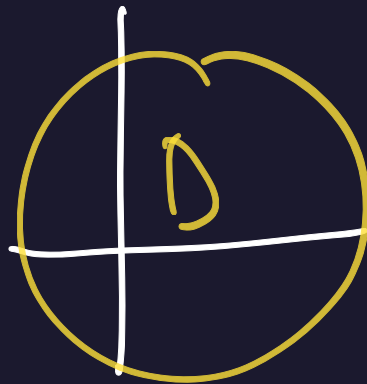
Surfaces Area of a Graph (and Length of a Graph)

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA$$

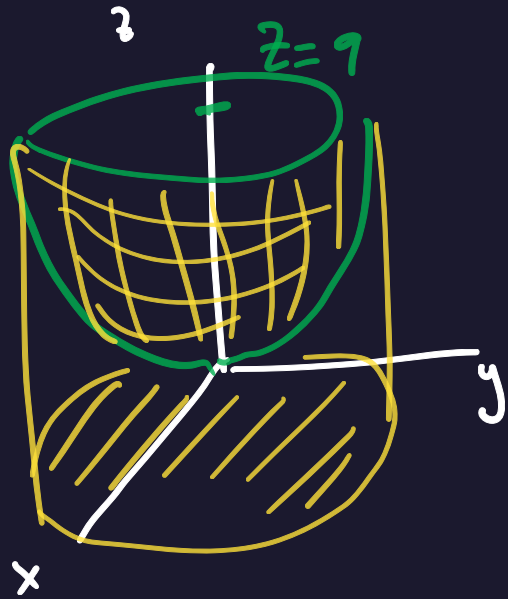
$$z = f(x, y)$$



$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$



Example: Compute the surface area of the paraboloid
 $z = x^2 + y^2$ below the plane $z = 9$.



$$z = \boxed{9 = x^2 + y^2} \leftarrow f(x, y) = x^2 + y^2$$

circle of radius 3

$$f_x = 2x$$

$$f_y = 2y$$

$$D = \{(x, y) : x^2 + y^2 \leq 9\}$$

$$r(u, v) = \{(u, v, u^2 + v^2) : u^2 + v^2 \leq 9\}$$

$$A(S) = \iint_D \sqrt{1 + (f_x)^2 + (f_y)^2} \, dA = \iint_D \sqrt{1 + 4x^2 + 4y^2} \, dA$$

$\sqrt{1 + 4(x^2 + y^2)}$

Example: Compute the surface area of the paraboloid
 $z = x^2 + y^2$ below the plane $z = 9$.

$$\iint_D \sqrt{1+4(x^2+y^2)} \, dA$$

$$x = r \cos \theta$$

$$0 \leq \theta \leq 2\pi$$

$$y = r \sin \theta$$

$$0 \leq r \leq 3$$

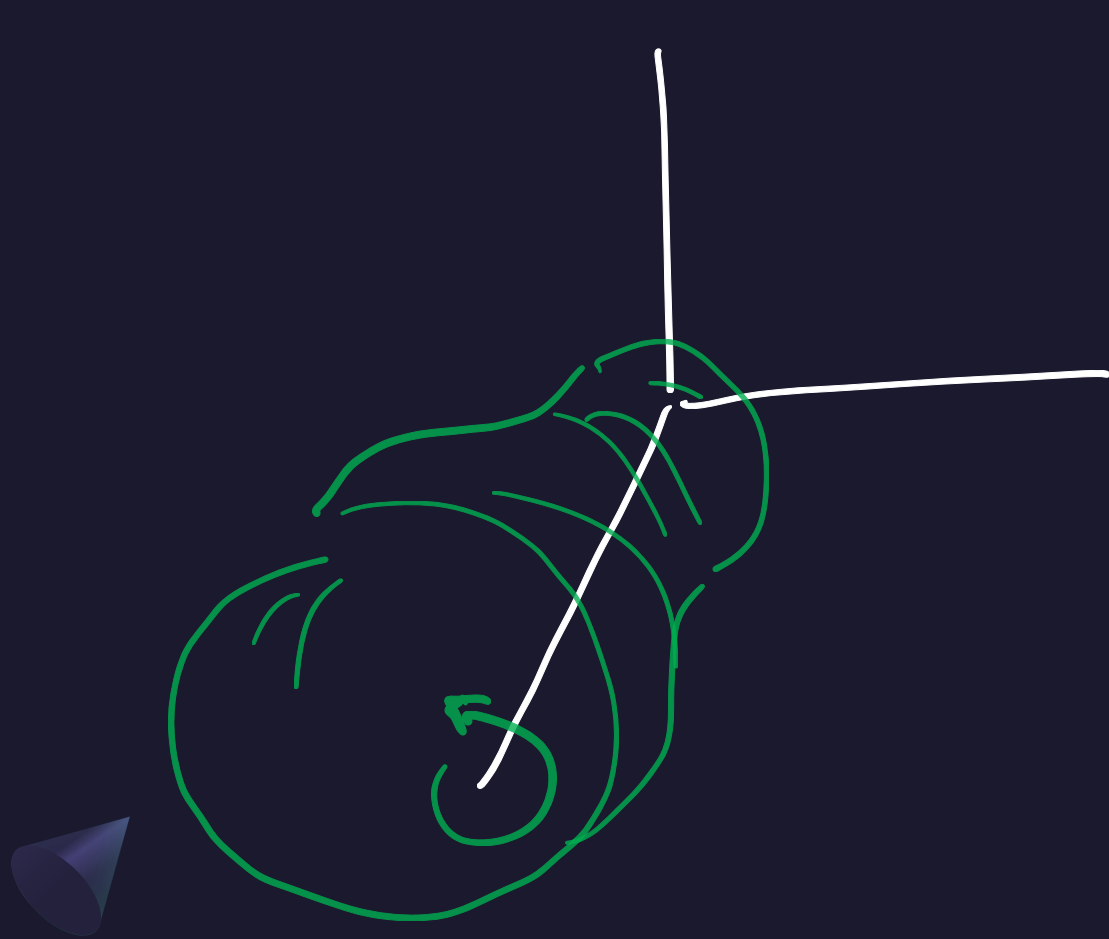
$$dA = r \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^3 \sqrt{1+4r^2} \cdot r \, dr \, d\theta = \underbrace{\int_0^{2\pi} d\theta}_{2\pi} \cdot \int_0^3 \sqrt{1+4r^2} \, dr$$

$$= 2\pi \int_0^3 r \sqrt{1+4r^2} \, dr = \left[\begin{array}{l} 1+4r^2 = u \\ du = 8r \, dr \\ r \, dr = \frac{1}{8} du \end{array} \right] = 2\pi \int_1^{37} \frac{1}{8} \sqrt{u} \, du = \frac{\pi}{6} (37^{3/2} - 1)$$

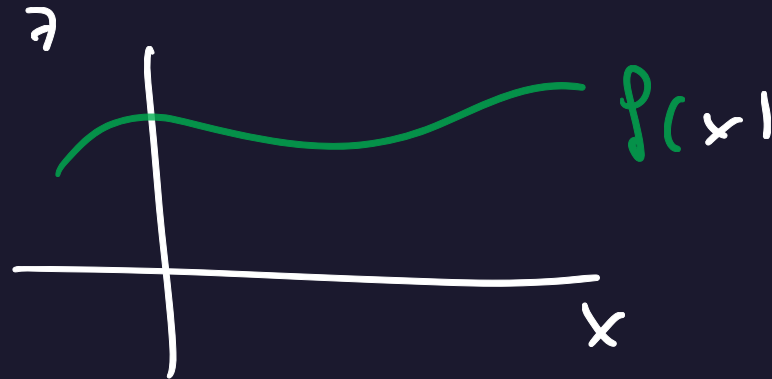
$r=0 \quad r=3$
 $u=1 \quad u=37$

Surfaces Area of a Surface of Revolution



$$r(u, v) = (u, f(u) \cos v, f(u) \sin v)$$

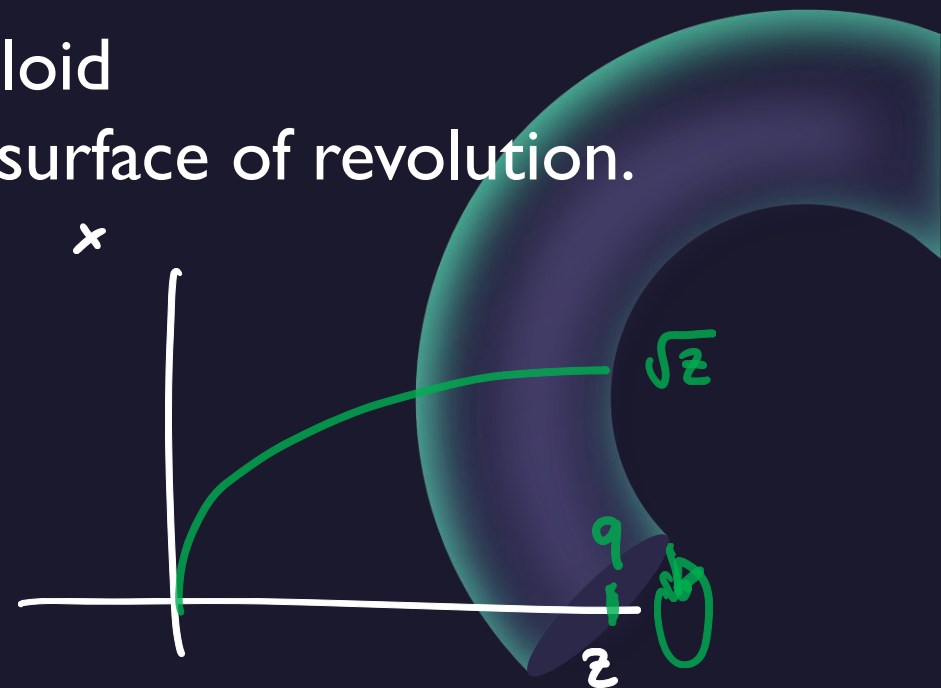
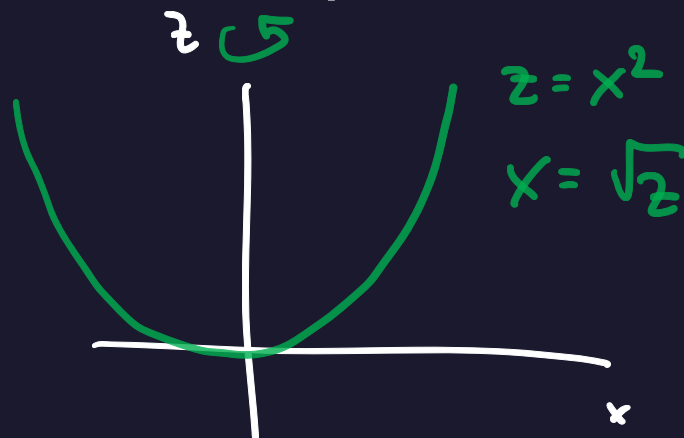
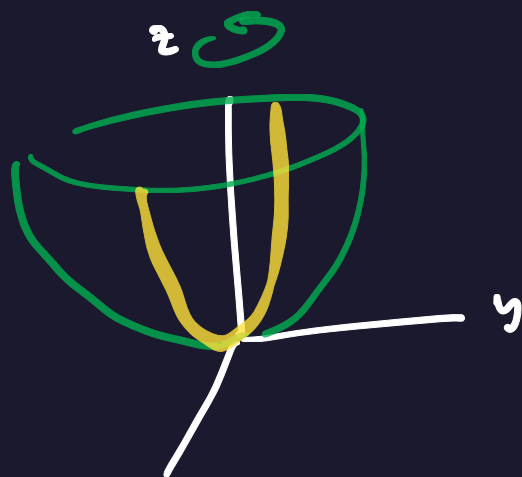
$$r_u \quad r_v \quad \dots \quad |r_u \times r_v|$$



Surfaces Area of a Surface of Revolution

$$\begin{aligned} A &= \iint_D |\mathbf{r}_x \times \mathbf{r}_\theta| \, dA \\ &= \int_0^{2\pi} \int_a^b f(x) \sqrt{1 + [f'(x)]^2} \, dx \, d\theta \\ &= 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} \, dx \end{aligned}$$

Example: Compute the surface area of the paraboloid $z = x^2 + y^2$ below the plane $z = 9$, as a surface of revolution.



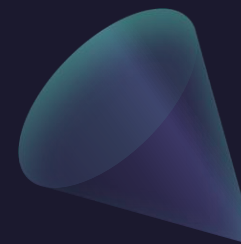
$$A(S) = \int_0^{2\pi} \int_0^9 \sqrt{z} \cdot \sqrt{1 + \frac{1}{4z}} \, dz \, d\theta$$

$$= \int_0^{2\pi} \int_0^9 \sqrt{z + \frac{1}{4}} \, dz \, d\theta$$

$$x = \sqrt{z}$$

$$x' = \frac{1}{2\sqrt{z}}$$

$$(x')^2 = \frac{1}{4z}$$



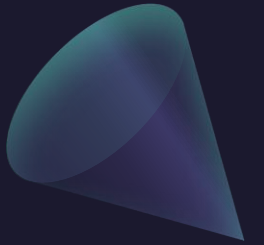
Example: Compute the surface area of the paraboloid
 $z = x^2 + y^2$ below the plane $z = 9$, as a surface of revolution.

$$A(S) = \int_0^{2\pi} \int_0^9 \sqrt{z + \frac{1}{4}} \, dz \, d\theta = \underbrace{\int_0^{2\pi} d\theta}_{2\pi} \cdot \int_0^9 \sqrt{z + \frac{1}{4}} \, dz$$

$$= 2\pi \int_0^9 \sqrt{z + \frac{1}{4}} \, dz = \left[\begin{array}{l} u = z + \frac{1}{4} \\ du = dz \end{array} \right] = 2\pi \int_{1/4}^{37/4} u^{1/2} \, du = \frac{\pi}{6} (37^{3/2} - 1)$$

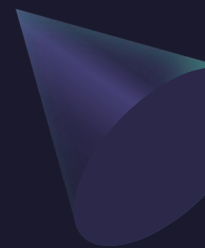
u	z
$1/4$	0
$37/4$	9

Questions?





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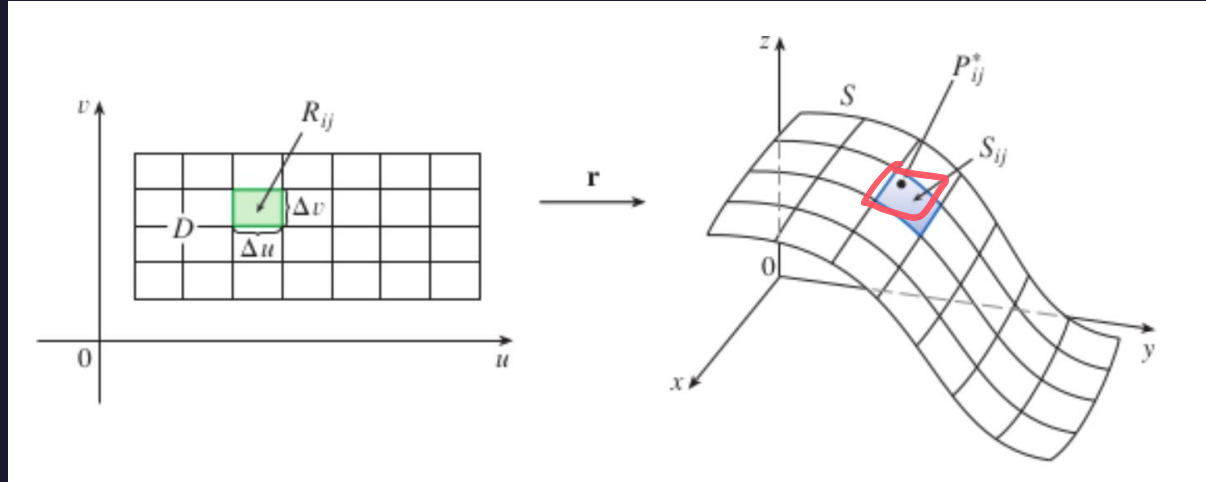
Instructor: Álvaro Lozano-Robledo

Surface Integrals



Integrals on Parametrized Surfaces

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$



$$\iint_S f(x, y, z) \, dS = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(P_{ij}^*) \, \Delta S_{ij}$$

$$\iint_S f(x, y, z) \, dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| \, dA$$

Example: Compute the surface integral $\iint_S x^2 dS$ where S is the unit sphere.

$$x^2 + y^2 + z^2 = 1$$

$$\uparrow f(x, y, z)$$



$$r(\phi, \theta) = (\sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi)$$
$$\begin{cases} 0 \leq \phi \leq \pi \\ 0 \leq \theta \leq 2\pi \end{cases}$$

$$f(r(\phi, \theta)) = x^2 = (\sin \phi \cos \theta)^2 = \sin^2 \phi \cdot \cos^2 \theta$$

$$|r_\phi \times r_\theta| = R^2 \sin \phi = \sin \phi$$
$$\uparrow R=1$$

$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$

Example: Compute the surface integral $\iint_S x^2 dS$ where S is the unit sphere.

$$\begin{aligned}
 \iint_S x^2 dS &= \int_0^{2\pi} \int_0^\pi \sin^2 \phi \cos^2 \theta \sin \phi d\phi d\theta \\
 &= \int_0^{2\pi} \underbrace{\cos^2 \theta d\theta}_{\frac{1 + \cos(2\theta)}{2}} \cdot \int_0^\pi \underbrace{\sin^3 \phi d\phi}_{\begin{aligned} &\sin \phi \sin^2 \phi \\ &= \sin \phi (1 - \cos^2 \phi) \\ &= \sin \phi - \sin \phi \cos^2 \phi \end{aligned}} d\phi = \pi \cdot \frac{4}{3} \\
 &= \boxed{\frac{4\pi}{3}}
 \end{aligned}$$

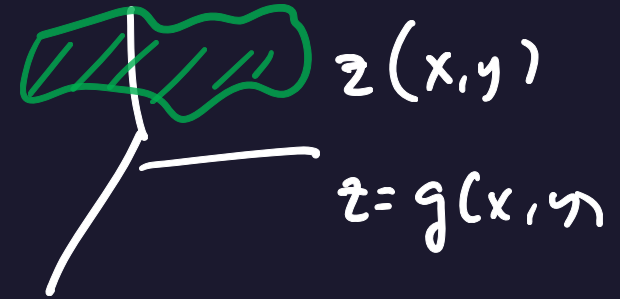
$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$

Example: Compute the surface integral $\iint_S x^2 dS$ where S is the unit sphere.

$$\begin{aligned}\iint_S x^2 dS &= \iint_D (\sin \phi \cos \theta)^2 |\mathbf{r}_\phi \times \mathbf{r}_\theta| dA \\&= \int_0^{2\pi} \int_0^\pi \sin^2 \phi \cos^2 \theta \sin \phi d\phi d\theta = \int_0^{2\pi} \cos^2 \theta d\theta \int_0^\pi \sin^3 \phi d\phi \\&= \int_0^{2\pi} \frac{1}{2}(1 + \cos 2\theta) d\theta \int_0^\pi (\sin \phi - \sin \phi \cos^2 \phi) d\phi \\&= \frac{1}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{2\pi} \left[-\cos \phi + \frac{1}{3} \cos^3 \phi \right]_0^\pi = \frac{4\pi}{3}\end{aligned}$$

Integrals on Graphs of Functions

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$



$$\iint_S f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| dA$$

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA$$

Example: Compute the surface integral $\iint_S y \, dS$ where S is the surface $z = x + y^2$, $0 \leq x \leq 1$, $0 \leq y \leq 2$.

$$f(x, y, z) = y$$

$$\mathbf{r}(u, v) = (u, v, u + v^2), \quad \begin{cases} 0 \leq u \leq 1 \\ 0 \leq v \leq 2 \end{cases}$$

$$g(x, y) = x + y^2$$

$$\sqrt{1 + (g_x)^2 + (g_y)^2} = \sqrt{1 + 1^2 + (2y)^2} = \sqrt{2 + 4y^2}$$

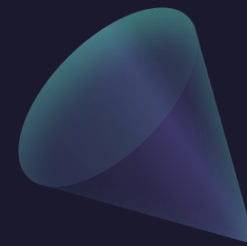
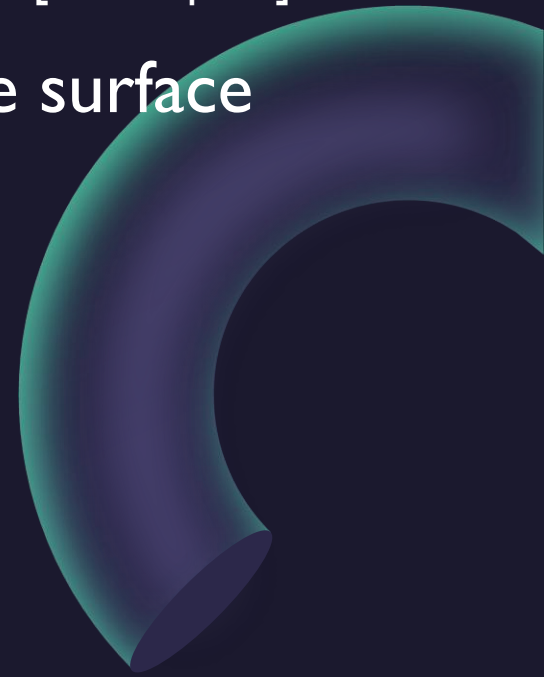
$$\frac{\partial g}{\partial x} = 1$$

$$\frac{\partial g}{\partial y} = 2y$$

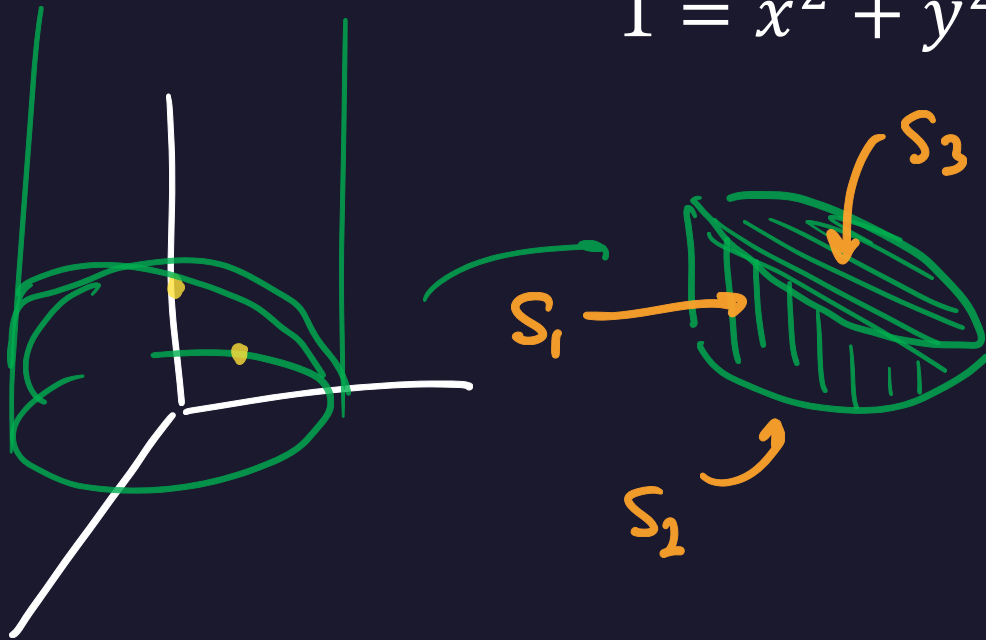
$$\iint_S y \, dS = \int_0^1 \int_0^2 v \cdot \sqrt{2 + 4v^2} \, dv \, du$$

Example: Compute the surface integral $\iint_S y \, dS$ where S is the surface $z = x + y^2$, $0 \leq x \leq 1$, $0 \leq y \leq 2$.

$$\int_0^1 du \cdot \int_0^2 v \cdot \sqrt{2+4v^2} \, dv = \left[\begin{array}{l} t = 2+4v^2 \\ dt = 8v \, dv \end{array} \right]$$
$$= \int_2^{18} \sqrt{t} \cdot \frac{1}{8} \, dt = \boxed{\frac{13\sqrt{2}}{3}}$$



Example: Compute the surface integral $\iint_S z \, dS$ where S is the surface whose sides are given by the cylinder and functions:
 $1 = x^2 + y^2$, $z = 0$, $z = 1 + x$.



$$\boxed{S_1} \quad x^2 + y^2 = 1$$

$$\mathbf{r}(\theta, z) = (\cos \theta, \sin \theta, z)$$

$$z = 1 + x = 1 + \cos \theta$$

$$\begin{cases} 0 \leq \theta \leq 2\pi \\ 0 \leq z \leq 1 + \cos \theta \end{cases}$$

$$0 \leq z \leq 1 + \cos \theta$$

$$\mathbf{r}(\theta, z) = (\cos \theta, \sin \theta, z)$$

$$\begin{cases} 0 \leq \theta \leq 2\pi \\ 0 \leq z \leq 1 + \cos \theta \end{cases}$$

$$\mathbf{r}_\theta = (-\sin \theta, \cos \theta, 0)$$

$$\mathbf{r}_z = (0, 0, 1)$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = (\cos \theta, \sin \theta, 0)$$

$|\cdot| = 1$
 $\downarrow$
 $\mathbf{r}_\theta \times \mathbf{r}_z$

Example: Compute the surface integral $\iint_S z \, dS$ where S is the surface whose sides are given by the cylinder and functions:

$$1 = x^2 + y^2, \quad z = 0, \quad z = 1 + x.$$

$$\iint_{S_1} z \, dS = \int_0^{2\pi} \int_0^{1+\cos\theta} z \, dz \, d\theta = \dots = \boxed{\frac{3\pi}{2}} \quad \boxed{\frac{3\pi}{2} + \sqrt{2}\pi}$$

$\boxed{S_2}$ ~~$z=0$~~ $r(u, v) = (r \cos \theta, r \sin \theta, 0) \quad \begin{cases} 0 \leq \theta \leq 2\pi \\ 0 \leq r \leq 1 \end{cases}$

$$\iint_{S_2} z \, dS = \iint_{S_2} 0 \cdot dS = \boxed{0}$$

$\boxed{S_3}$

graph of
 $z = 1 + x$

$$\begin{aligned} \iint_{S_3} z \, dS &= \iint_D (1+x) \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \, dA \\ &= \int_0^{2\pi} \int_0^1 (1+r \cos \theta) \sqrt{2} \, dr \, d\theta \end{aligned}$$

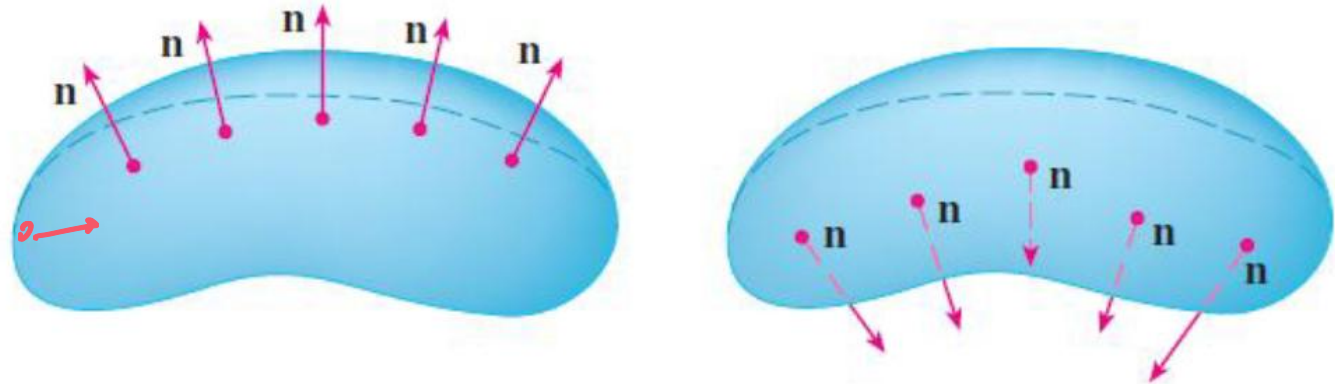
$\boxed{\sqrt{2}\pi}$

Orientation

- ▶ For surface integrals of a vector field $\vec{F}$ over a surface S , the orientation of the surface must be specified, just like the line integrals. The orientation of a surface is specified via its normal vectors at any point.
- ▶ For a closed surface S , we define **the positive orientation** of S to mean that all normal vectors to S point outward (with negative meaning normal vectors point inward instead). If it is impossible to have all normal vectors always point either outward or inward, then the surface S is non-orientable.

FIGURE 7

The two orientations
of an orientable surface



Orientation

Möbius band



- ▶ To define a surface integral of a vector field, we only want to count the portion of $\vec{F}$ that is normal to the surface S at any point. Hence, the dot product again plays a role in the computation.
- ▶ Then the **surface integral** of $\vec{F}$ over the surface S is defined as

$$\begin{aligned}\iint_S \vec{F} \cdot d\vec{S} &= \iint_S \vec{F} \cdot \vec{n} \, dS \\ &= \iint_S \vec{F} \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} dS \\ &= \iint_D \left[\vec{F} \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \right] |\vec{r}_u \times \vec{r}_v| dA \\ &= \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) dA.\end{aligned}$$

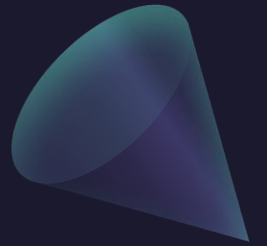
length 1 and normal

This is often called the **flux integral** as it measures the flux of $\vec{F}$ across the surface S .

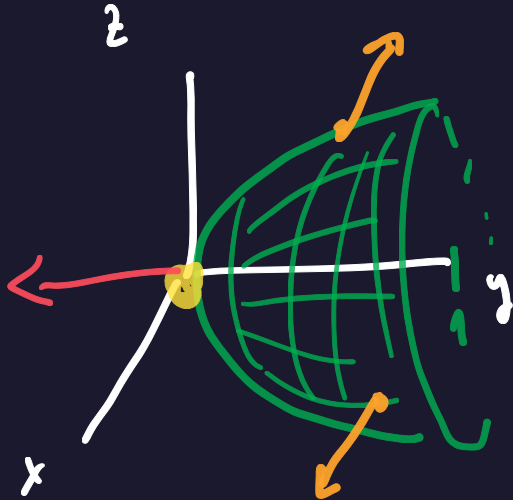
- ▶ Recall and compare it with the line integral of vector fields:

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt.$$

Orientation



Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2$, $0 \leq y \leq 1$.
(Assume S is positively oriented.)



$$D = \{x^2 + z^2 \leq 1\}$$

$$r(u, v) = (u, u^2 + v^2, v) \quad u^2 + v^2 \leq 1$$

$$r_u = (1, 2u, 0)$$

$$r_v = (0, 2v, 1)$$

$$(u, v) = (0, 0)$$

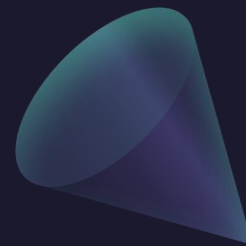
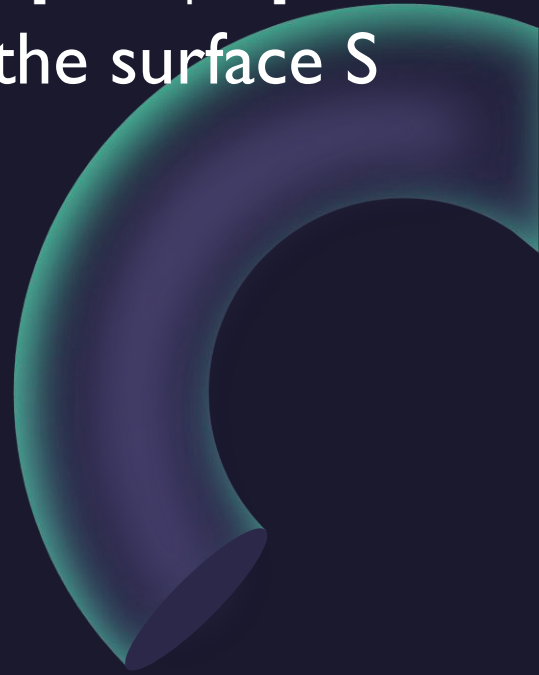
$$r(0, 0) = (0, 0, 0)$$

$$n = (0, -1, 0) \quad \checkmark$$

pos. oriented !!

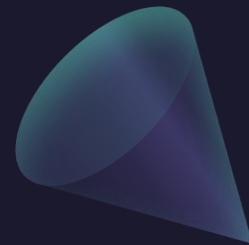
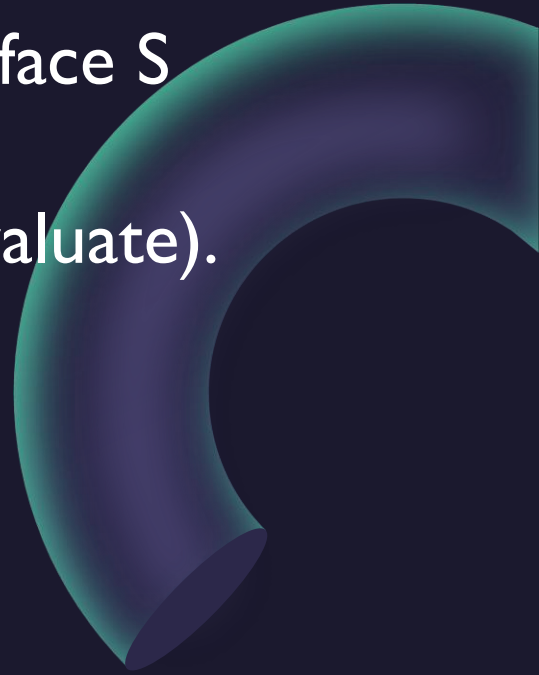
$$n = r_u \times r_v = \begin{vmatrix} i & j & k \\ 1 & 2u & 0 \\ 0 & 2v & 1 \end{vmatrix} = (2u, -1, 2v)$$

Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2$, $0 \leq y \leq 1$.
(Assume S is positively oriented.)



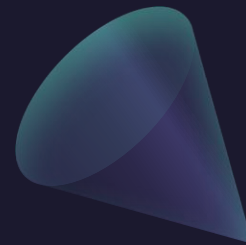
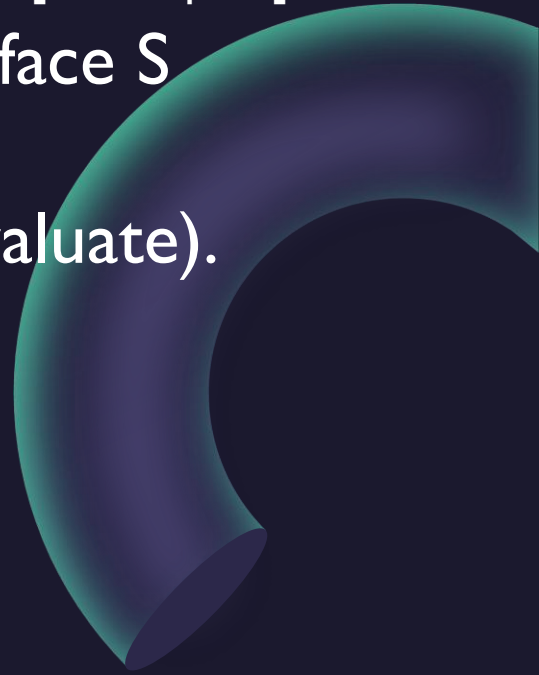
Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$. Assume S is positively oriented. Set up (but do not evaluate).

$$\iint_S \operatorname{curl}(F) \cdot dS$$

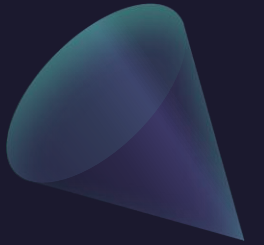


Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$. Assume S is positively oriented. Set up (but do not evaluate).

$$\iint_S \operatorname{curl}(F) \cdot dS$$



Questions?



Thank you

Until next time.

